

Chapter 147 — Surgery of the Adrenal Glands

HISTORY OF ADRENAL SURGERY

- **First printed description of the adrenal glands:** Eustachius, mid-1500s. More than three centuries passed before the first surgical excision.
- **1889, Thornton — first reported adrenal excision:** performed as an "abdominal nephrectomy" for a very large tumour that proved to be adrenal and hormonally active, through a T-shaped subcostal (chevron or hemi-chevron) incision.
- **1905 — first adrenalectomy for a preoperative diagnosis of adrenal pathology** ("enlarged" glands). **1934 — first series of adrenalectomies**, all for Cushing syndrome, with mixed pathology (carcinoma, hyperplasia, adenoma and normal glands).
- **Preoperative steroid cover for Cushing syndrome was the greatest early advance:** operative mortality, previously substantial even after subtotal excision, fell to zero once steroids were given preoperatively (reported 1949).
- **Imaging refined selection:** ultrasonography (US), computed tomography (CT), magnetic resonance imaging (MRI) and [^{131}I]-meta-iodobenzylguanidine (MIBG) scintigraphy separated patients who benefit from surgery from those who do not.
- **Access migrated cephalad** as the goal shifted from metabolic control to tumour excision: subcostal → flank with 11th or 12th rib resection → **1932: transpleural, transdiaphragmatic long posterior intercostal incision**, forerunner of the thoracoabdominal incision — virtually unparalleled exposure at the price of thoracic complications and a painful, protracted recovery. Anterior, posterior, flank, transperitoneal and retroperitoneal approaches all retain advocates and critics.
- **Open surgery was the gold standard for about a century**, until 1992. Since then, surgical adrenal disease has been synonymous with minimally invasive surgery (MIS), with very few exceptions.

MINIMALLY INVASIVE MILESTONES

Year	Milestone
1992	First laparoscopic adrenalectomy — transperitoneal, Canada (Gagner)
1999	First robot-assisted adrenalectomies — two European centres, AESOP platform (now obsolete)
2002	First da Vinci adrenalectomy — United States urologists (one pheochromocytoma, one benign adenoma)
2008	First single-port adrenalectomy — one skin incision, three fascial ports, no dedicated device; led to laparoendoscopic single-site surgery (LESS) devices
2018 / 2020 / 2022	Single-port (SP) da Vinci approved by the Food and Drug Administration (FDA) / first SP procedures / first reported

- **Outlook:** numerous robotic platforms now in use; artificial intelligence (AI), augmented reality (AR) and three-dimensional simulation anticipated.

Concept — the history is a double migration — the incision moved cephalad as the aim moved from metabolic control to tumour excision, then the approach became minimally invasive — but the first advance that made adrenalectomy safe was hormonal (steroid cover), not technical.

BRIEF ANATOMY OF ADRENAL GLAND

- **Paired retroperitoneal (RP) organs within Gerota's fascia**, on the superomedial aspect of each kidney; 3–5 cm transverse, about 5 g each.

- **Cortex:** bright yellow-orange, finely granular, firm — distinguishes it from surrounding fat. **Medulla:** dark red, more friable, not visible surgically.

RIGHT VERSUS LEFT ADRENAL GLAND

Feature	Right	Left
Shape	Pyramidal	Crescentic
Position	Above upper pole of right kidney, between liver and diaphragm	Superomedial to left kidney, lateral to aorta
Relations	Bare area of the liver , separated only by a thin fibrous layer (fused to liver in ~10%); duodenum anteromedially; inferior vena cava (IVC) medially; diaphragm posteriorly; kidney inferiorly	Lesser-sac peritoneum over superior half of anterior surface (separates it from gastric cardia and spleen); no peritoneum inferiorly , where pancreas body and splenic artery and vein lie anterior to the gland; medially coeliac ganglion, left inferior phrenic and left gastric arteries on the crus
Mobility	Less mobile	More mobile, more accessible
Vein	Single, <1 cm ; enters posterior IVC at 45° ; working length often gained only late in mobilisation	Single, longer ; runs inferiorly, joined by inferior phrenic vein, into left renal vein
Venous variants	Into right hepatic vein or right renal vein	Into left phrenic vein first; or across aorta directly into IVC

- **Arterial supply — three arteries to each gland:** superior adrenal (from inferior phrenic artery), middle adrenal (from aorta), inferior adrenal (from renal artery). Accessory vessels from intercostal arteries and the left gonadal artery.
- **Lymphatics:** lateral aortic chain from diaphragm to ipsilateral renal artery; may end in thoracic duct or, after piercing the crura, posterior mediastinal nodes.

Concept — *three arteries but one vein: adrenalectomy is organised around the vein, and the right vein's shortness and direct caval entry make the right the more hazardous side.*

INDICATIONS FOR ADRENAL SURGERY

- **Guidelines:** American Association of Endocrine Surgeons (AAES) adrenalectomy guideline, 2022; American Association of Clinical Endocrinologists (AACE)/AAES recommendations, 2009 — still widely practised.

Adrenal Masses

- **Incidentalomas occur in about 4%** (range 1–8.7%) of the population.
- **Hormonally active tumours** (phaeochromocytoma, aldosteronoma) → **excise regardless of size. All suspected phaeochromocytomas and adrenocortical carcinomas (ACCs) → excise. <1 cm and non-functioning → observe.**
- **Solid enhancing lesion may be a metastasis** (primaries: lung, breast, melanoma, kidney, thyroid, colon — ranked order under Surgery for Metastatic Lesions). With a known primary, strongly consider removal when it is the **solitary site of spread** or will improve quality or quantity of life; low MIS morbidity has made surgeons more aggressive, with encouraging results.

Hormonally Active Lesions

- **Adrenalectomy or partial adrenalectomy when systemic manifestations cannot be controlled medically.**

BOX 147.1 INDICATIONS FOR ADRENALECTOMY

Category	Indication
Functional	Aldosterone-secreting adenoma (Conn syndrome); cortisol-secreting adenoma (Cushing syndrome); pituitary-dependent Cushing disease after failed trans-sphenoidal surgery ; bilateral adrenal hyperplasia; pheochromocytoma; androgen- or oestrogen-producing tumour causing virilisation or feminisation
Non-functional	Histologically confirmed ACC; symptomatic masses (cysts, myelolipomas); incidentalomas meeting size or imaging criteria; solitary metastasis from a non-adrenal primary
Size criteria	≥4 cm; >2 cm growing ≥1 cm in 12 months ; <1 cm growing >6–12 mm in 6–12 months (may be offered surgery)
Imaging criteria	>10 Hounsfield units (HU) on non-contrast CT ; contrast CT washout <50% with delayed attenuation >35 HU ; MRI chemical shift not consistent with adenoma; ¹⁸ F-fluorodeoxyglucose positron emission tomography (FDG-PET) uptake present or greater than liver

- **What the box tests:** two independent routes to surgery — function (any size) and malignant potential (size, growth, imaging phenotype).
- **Trap:** "bilateral adrenal hyperplasia" appears as a functional indication, yet bilateral hyperplasia causing hyperaldosteronism is **not** surgically treatable — hence adrenal venous sampling before adrenalectomy for aldosteronism.

Concept — *two gates lead to adrenalectomy: hormonal function, which mandates excision whatever the size, and malignant potential, judged by size, growth and imaging phenotype.*

CONTRAINDICATIONS TO ADRENAL SURGERY

Contraindication (correct before operating)	Relative
Uncontrolled hypertension; uncontrolled coagulopathy; untreated bacterial peritonitis (for the transperitoneal route)	Pregnancy

Uncontrolled Hypertension

- **Blood pressure (BP) control is a priority before, during and after surgery** — prevents myocardial events, cerebrovascular accidents and renal barotrauma.
- **Unopposed β-blockade is highly contraindicated** until α-blockade is titrated and BP stable: loss of β₂-mediated vasodilatation leaves catecholamine-driven α-vasoconstriction unopposed → hypertensive crisis, sometimes fatal.
- **Other hypertensive adrenal conditions** often need combinations — angiotensin-converting enzyme inhibitors (ACE-Is), angiotensin II receptor blockers (ARBs), calcium-channel blockers (CCBs), diuretics, β-blockers — to reduce lability and spikes.

PREOPERATIVE BP CONTROL IN PHAEOCHROMOCYTOMA

Agent	Class	Place
Phenoxybenzamine	Non-selective, non-competitive (irreversible) α-blocker	Preferred
Prazosin, doxazosin, terazosin	Selective α ₁ -blockers	Acceptable; less desirable
CCBs	—	Reasonable alternative
Metyrosine	—	Alternative; more adverse effects with prolonged use

Agent	Class	Place
β -blockers	—	Only after adequate α -blockade

Uncontrolled Coagulopathy

- **Contraindication to most surgery;** given adrenal vascularity, normalise platelets, bleeding time, prothrombin time (PT), partial thromboplastin time (PTT) and international normalised ratio (INR) first.
- **Low-dose aspirin (81 mg) does not increase intra- or postoperative bleeding;** do not discourage it where benefit outweighs risk.

Untreated Bacterial Peritonitis

- **Avoid transperitoneal surgery;** an RP approach may be used, with great care not to breach the peritoneum.
- **Insufflation pressure** (trans- or retroperitoneal) can exceed central and systemic venous pressure → abdomino-venous or abdomino-lymphatic backflow → bacteraemia, sepsis. Eradicate infection first.

Pregnancy (Relative)

- **Defer elective adrenal surgery until after delivery.** Pheochromocytoma, cyst rupture or tumour haemorrhage may compel surgery during pregnancy.
- **Historical versus current view:** second trimester was held safest (first → spontaneous abortion; third → preterm labour). American College of Obstetricians and Gynecologists (ACOG) and Society of American Gastrointestinal and Endoscopic Surgeons (SAGES) now advise that **surgically necessary treatment is not delayed, regardless of trimester.**
- **Open and MIS both acceptable;** obstetric service aware, available and present.

Past Medical and Surgical History Relevant to Adrenal Surgery

- **Clues to prior or ongoing adrenal disease:** hypertension (especially uncontrolled), easy bruising, palpitations, headaches, hirsutism, stunted growth, anisomelia.
- **Prior nephrectomy may have removed the ipsilateral adrenal** — contralateral adrenalectomy is then dangerous if unrecognised. Flank scars are a clue; review cross-sectional imaging and consider serum testing for intact adrenal tissue.
- **Pneumoperitoneum:** pulmonary disease, cardiac insufficiency, ocular disease, morbid obesity or prior steroids raise inspiratory pressures and lung-complication risk; gasless laparoscopy is an option, not widely adopted.
- **Prior abdominal surgery** → adhesions, prolonged or impossible transperitoneal MIS, bowel injury → **RP access may be superior.**
- **Open surgery:** large, painful incisions; steroids impair healing; acute pain hard to control.

Concept — *each contraindication is a correctable state — blood pressure, clotting, infection, gestation — so it contraindicates operating now, not adrenalectomy itself.*

THE INCIDENTALOMA

- **Definition:** adrenal mass ≥ 1 cm found on imaging for an unrelated reason; usually < 4 cm. Management is multidisciplinary (endocrinology, radiology, surgery).
- **Frequency:** about 9% at autopsy; about 5% of abdominal imaging studies.
- **Two questions — functional? malignant?** ACC in about 2%; autonomous hormone secretion in up to 20%.

Evaluation for Hormone Hypersecretion

- **History and examination first:** weight gain, centripetal obesity, bruising, hypertension, diabetes, virilisation, proximal muscle weakness, fatigue.

BIOCHEMICAL SCREENING OF THE INCIDENTALOMA

Excess	Test whom	Screen	Positive / next	Trap
Cortisol	All	1-mg overnight dexamethasone suppression test (highest sensitivity for subclinical Cushing); late-night salivary cortisol; 24-hour urinary free cortisol (standard, cumbersome)	—	Never a spot serum cortisol — false negatives
Catecholamines (phaeochromocytoma in 4–5%)	All	Plasma free metanephrines and normetanephrines , or 24-hour urinary total metanephrines and fractionated catecholamines	>2× upper limit of normal	Metanephrines preferred — continuous, not episodic, production
Aldosterone	With refractory hypertension	Morning plasma aldosterone-to-renin ratio (ARR)	ARR >20:1 with aldosterone >20 ng/dL → salt-loading 24-hour urine to confirm → adrenal venous sampling before surgery	Up to 40% normokalaemic — never rely on spot potassium
Sex hormones	Mass ≥4 cm and/or virilisation or feminisation	Serum dehydroepiandrosterone (DHEA), 17-ketosteroids; testosterone in women, 17β-oestradiol in men	—	ACC is only ~2% of incidentalomas, but secretes sex hormones in most cases

- **Conn syndrome** (aldosteronism from a single nodule): classic hypertension, hypokalaemia, alkalosis; a **surgically curable cause of hypertension**. Bilateral hyperplasia also causes aldosteronism but is not surgically treatable — the reason for adrenal venous sampling.

Radiographic Evaluation

- **Hormonally inert → non-contrast CT** for size and characteristics.

Benign features	Suspicious features
≤4 cm, ≤10 HU, homogeneous → minimal malignant risk; repeat adrenal-protocol CT at 12 months	>4 cm, heterogeneous, hypervascular, >10 HU → chemical-shift MRI, FDG-PET-CT or ¹²³ I-MIBG single-photon emission CT (SPECT-CT); biopsy in some

- **Myelolipoma on imaging** (macroscopic fat, negative attenuation ≤ −20 HU): biochemistry may be unnecessary, but test if clinical features suggest excess — 10–15% may be hormonally functional.

Management

- **<4 cm and non-functional → surveillance imaging.** **>4 cm or functional → surgical removal.** Age and health status also weigh.

Concept — *biochemistry precedes imaging because function decides surgery whatever the size; imaging only grades malignant risk in the hormonally inert mass.*

SURGICAL APPROACH TO THE ADRENAL

- **Choice depends on** disease, tumour size, patient factors, surgeon experience, equipment and preference; laterality poses distinct challenges.

BOX 147.2 SURGICAL APPROACHES TO THE ADRENAL GLAND

	Open	Minimally invasive surgery (MIS)
Transperitoneal	Subcostal; midline; chevron; extended chevron	Laparoscopic; hand-assisted; gasless; robot-assisted; single-site
Retroperitoneal	Flank; thoracoabdominal; dorsal lumbotomy	Laparoscopic; robot-assisted; single-site

Transperitoneal

- **Oldest documented approach.** "You can never be too high" — the adrenal's superior position limits exposure.
- **Midline:** early, excellent vascular control and view of great vessels; no muscle cut → less pain, better healing, fewer hernias and dehiscences. **Subcostal/chevron:** crosses several muscles → more pain, but "higher" exposure; excellent for smaller tumours and hormonally active adenomas or hyperplasia.
- **Layers — midline:** skin → subcutaneous tissue → linea alba → transversalis fascia → preperitoneal fat → peritoneum. **Subcostal/chevron:** skin → subcutaneous tissue → anterior rectus sheath and rectus medially; external oblique, internal oblique, transversus abdominis laterally → transversalis fascia → preperitoneal fat → peritoneum.
- **Fig. 147.1 — arcuate line:** above it the rectus lies within anterior and posterior sheaths (internal oblique aponeurosis splits); below it all aponeuroses pass anterior, leaving rectus on transversalis fascia.
- **Closure:** laterally two fascial layers (transversus abdominis; external oblique), converging to one layer medially. Peritoneal closure is debated — conflicting evidence on hernia prevention.
- **Ports:** laparoscopy — all-midline, subcostal, "L" or reverse-"L"; **no configuration proven superior.** Robotic — **subcostal most common:** excellent exposure, no instrument collision (Figs 147.2, 147.3). Single-site — one medium incision, gel platform or SP robot.

Right Adrenal

Sequence: elevate liver (no capsular tears), incise peritoneal attachments → **Kocher manoeuvre** (medial reflection of second part of duodenum) exposes IVC → dissect along IVC to **right renal vein insertion — inferior limit** → continue cephalad with liver retraction to superior adrenal and **adrenal vein** → isolate and control vein early (clips or thermal energy), **warning anaesthesia** → downward renal traction reveals superior adrenal (electrocautery) → elevate adrenal packet to **psoas — posterior limit** → lateral dissection to body wall → free from liver, kidney and IVC → extraction bag, tumour intact.

- **Functional tumour:** open communication with anaesthesia around vein control prevents rapid hypotension and cardiovascular collapse. **Adrenal arteries are seldom seen** — electrocautery or thermal devices; brisk bleeding rare.

Left Adrenal

Sequence: unless transmesenteric, reflect colon along the **white line of Toldt**, staying outside Gerota's fascia (in obese patients, carry below iliac vessels) → elevate spleen carefully → identify **left renal vein** (or trace gonadal vein cephalad) → dissect superior border of renal vein to **adrenal vein** → clip or seal after warning anaesthesia → gland lies immediately posterior and superior → medial dissection with lateral renal traction exposes **splenic artery and pancreas** (no undue traction) → incise peritoneum as spleen elevated to expose superior adrenal, staying superior → lateral dissection to abdominal wall and diaphragm → free superior

renal border → divide **nexus of ganglia** superomedially and posteriorly (thermal energy or clips) → **psoas — posterior limit** → bag, specimen intact.

TRANSPERITONEAL RIGHT VERSUS LEFT

	Right	Left
First manoeuvre	Liver elevation + Kocher	Colon reflection along line of Toldt
Guiding vessel	IVC → right renal vein insertion	Left renal vein (or gonadal vein)
Structures at risk	Liver capsule, IVC, duodenum	Spleen, pancreas, splenic artery, diaphragm
Side-specific step	—	Division of ganglia nexus
Posterior limit	Psoas muscle (both)	

Retroperitoneal

Open Approach

Positioning. Lateral decubitus, affected side up; table flexed at costal margin; optional kidney rest; axillary roll; arms extended away, upper arm on padded elevated rest; lower leg bent, upper straight (Fig. 147.4).

Incision. Flank; divide latissimus dorsi, external oblique, internal oblique, transversus abdominis; reflect parietal pleura upward; resect **12th (occasionally 11th) rib**.

Rib resection: scrape anterior periosteum (periosteal elevator) → strip posterior periosteum from posterior and superior aspects, **protecting the neurovascular bundle in the groove on the posteroinferior border** (nerve injury → **flank bulge**) → divide distally with rib cutter → file edge; electrocautery and bone wax → enter lumbodorsal fascia, develop RP space bluntly with finger, sweep peritoneum off transversalis fascia → divide overlying muscles; fixed retraction.

Dissection — right. Medially over IVC, dividing posterior peritoneum → superiorly to adrenal vein; if needed, first control **short hepatic veins** between silk ties for **1–2 cm more superior exposure** (posterosuperior adrenal vein) → ligate adrenal vein between silk ties → superior extent between adrenal and liver → **adrenal–kidney attachment last**, as a handle for downward retraction.

Dissection — left. Medially to aorta → superiorly to left renal vein → along it to adrenal vein; ligate between silk ties → superior extent, protecting **pancreatic tail and splenic vessels** → adrenal–kidney attachment last. On both sides the multiple arteries are controlled with electrocautery devices.

Closing. Inspect for haemostasis (warm sterile water irrigation helps) → two-layer fascial closure, running bidirectional **looped polydioxanone 0 or #1**: deep — transversus abdominis and internal oblique; superficial — external oblique → skin staples or absorbable monofilament.

Concept — *whichever cavity is entered, the operation is the same: follow the renal vein to the adrenal vein, control the vein, and keep the adrenal–kidney attachment until last as a retraction handle.*

MINIMALLY INVASIVE SURGICAL APPROACHES

- **Laparoscopic and robot-assisted** approaches both suit adrenal pathology; lateral or prone positioning; instruments by surgeon preference.
- **RP MIS versus transperitoneal MIS:** avoids bowel and viscera — preferred after prior abdominal surgery. **RP MIS versus open RP:** less pain, shorter stay, though regional anaesthesia narrows the gap.
- **Robot particularly suits the RP approach** — small working space, angled instruments.

Minimally Invasive Surgery Retroperitoneal Adrenalectomy Technique

Positioning. As for the open retroperitoneal approach.

Access and ports: 1-cm transverse incision near tip of 12th rib, beneath 11th → divide or spread muscles to lumbodorsal fascia → open; enter RP space, confirm with finger → finger-dissect space from undersurface of 11th rib (superior), iliac crest (inferior), anterior psoas (posterior); sweep peritoneum anteromedially → RP balloon dissector inflated under vision → insufflate → two further ports **8 cm apart**: lateral just below line of 12th rib; medial subcostal, **3 cm below junction of 12th rib and paraspinous muscle** (Fig. 147.5; dotted circles mark alternative third-port sites).

Left Minimally Invasive Surgery Adrenalectomy Dissection

Sequence: medially along psoas and aorta → superiorly to renal vessels → **renal artery (posterior to vein) met first**; dissect circumferentially → **adrenal vein anterior and cephalad to superior border of renal artery** (Fig. 147.6) → divide with clips or vascular stapler (Fig. 147.7) → small arteries on inferior and medial surface: bipolar, divide → free circumferentially, **anterior and superior attachments last** for downward retraction.

Right Minimally Invasive Surgery Adrenalectomy Dissection

- **Medially along psoas and IVC** → superiorly beneath right renal hilum.
- **Right adrenal vein enters posterolateral IVC, superior to right renal hilum** (Fig. 147.8); shorter and possibly wider than left — **avulsion from IVC → significant haemorrhage**. Otherwise as for left.
- **Closure:** endoscopic retrieval bag through the 10-mm port; inspect bed; close ports.

Transperitoneal: first vessel met	Retroperitoneal: first vessel met
Renal vein (anterior)	Renal artery (posterior to vein)

Concept — *the retroperitoneoscopic view reverses the transperitoneal one — posterior structures (psoas, renal artery) come first — which is why it suits the previously operated abdomen and why its landmarks must be learnt afresh.*

SURGERY FOR PHAEOCHROMOCYTOMA

Preoperative Management and Coordination With Anaesthesia

- **Presentation:** tachycardia, hypertension, diaphoresis, headache, arrhythmia, left ventricular dysfunction, impaired glucose tolerance.
- **Work-up:** electrocardiogram, echocardiogram, assessment of hypertensive end-organ damage; plan jointly with anaesthesia — hypertensive spikes and tachyarrhythmias are expected.

Preparation: **α -blockade 2 weeks before surgery** — phenoxybenzamine (irreversible non-selective) or doxazosin (selective α_1) — titrated to normotension → **β -blockade only if tachycardia or arrhythmia** → **volume expansion** (hydration or high-salt diet), because chronically vasoconstricted patients become hypotensive once vasodilated.

Intraoperative Considerations

- **Invasive monitoring;** tell anaesthesia whenever the mass is being manipulated; pause manipulation during lability. **Ligate the adrenal vein early** if size allows.

INTRAOPERATIVE HAEMODYNAMIC CONTROL

Event	Agents / action
Hypertensive surge	Nitroprusside, nicardipine, nitroglycerin, phentolamine; pause manipulation
Tachyarrhythmia	Esmolol, lidocaine
Hypotension after vein ligation	Aggressive volume repletion $\pm$ α -adrenergic agonists

Postoperative Considerations

- **Invasive haemodynamic monitoring** of BP, heart rate and blood glucose.
- **Lifelong recurrence up to 16%** → regular long-term BP monitoring and biochemical screening.

Concept — *phaeochromocytoma surgery is a two-phase haemodynamic problem — catecholamine surge while the tumour is handled, catecholamine withdrawal once the vein is tied — and α -blockade with volume expansion prepares for both.*

SURGERY FOR MALIGNANCY

Adrenocortical Carcinoma

- **Only two primary adrenal malignancies in adults: ACC and phaeochromocytoma**; neuroblastoma is paediatric and not covered.
- **ACC**: rare, of cortical origin, **0.5–2 per million per year**.
- **Who operates**: European Society of Endocrinology (ESE) and European Network for the Study of Adrenal Tumours (ENSAT) — surgeon **>6 adrenalectomies per year, preferably ≥ 20** , with extensive oncological experience, and equally experienced anaesthesia and oncology teams.
- **How**: **open transperitoneal**, removing all visible tumour, with en-bloc adjacent organ resection as needed (liver, kidney, spleen, intestine, IVC, aorta, musculoskeletal soft tissue).
- **Intact specimen is essential** — rupture gives dismal results. Challenged by many MIS surgeons, but the National Comprehensive Cancer Network (NCCN) also supports open surgery.
- **Advanced-stage 5-year survival <15%**. Global recommendation: **open adrenalectomy with intact removal for suspected ACC**, especially larger tumours; multidisciplinary care (medical and radiation oncology, radiology, endocrinology, anaesthesiology, pathology) and tumour boards.

Surgery for Metastatic Lesions to the Adrenal

- **Metastases are the commonest malignant adrenal tumour, and the second commonest adrenal tumour after benign adenoma**: 2.8–6.7% of adrenal tumours; prevalence 0.5–3.8%; median size about 3 cm; mostly unilateral, about a quarter bilateral.

BOX 147.3 ORIGINS OF METASTATIC TUMOURS TO THE ADRENAL GLAND

Rank	Primary	Proportion
1	Lung	39.2%
2	Genitourinary (vast majority kidney)	27.6%
3	Gastrointestinal	13.6%
4	Lymphoma	5.0%
5	Skin (melanoma)	4.0%
6–9	Sarcoma; breast; neuroendocrine; thyroid	2.9%; 1.7%; 1.0%; 0.7%
—	Other	4.3%

- **Pattern and trap**: lung and kidney together account for about two-thirds; breast, often listed among common primaries, is actually uncommon.
- **Technique is that of routine adrenalectomy; the goal is eradication**. One of many sites or widespread disease → adrenalectomy may not help. **Adrenal is the only measurable disease → long-term benefit possible**.
- **Survival depends on primary**: renal-origin metastases do far better than lung, breast or melanoma.
- **Better prognosis**: renal origin, synchronous tumour, size <5 cm, symptomatic tumour.

- **Prior systemic chemotherapy or ipsilateral radiation** → tedious dissection around liver, spleen, bowel and great vessels; involve oncologists; colorectal, vascular or general surgeons may be needed.

Concept — *both malignant indications are judged by curative intent — for ACC, complete, intact, open resection; for metastasis, removal only when the adrenal is the sole disease.*

SURGERY FOR ALL OTHER TUMOURS

Surgery for Myelolipoma

- **Benign: mature adipose plus haematopoietic tissue**; usually asymptomatic and incidental; **macroscopic fat with variable calcification** on non-contrast imaging.
- **Frequency**: up to 0.4% at autopsy; **10–15% of incidentalomas**.
- **Presentation when symptomatic**: abdominal pain, **spontaneous retroperitoneal haemorrhage**.
- **Most need no treatment. Operate if >7 cm, symptomatic, or adverse imaging features** suggesting malignancy.
- **Still screen for pheochromocytoma and Cushing syndrome** — excludes pheochromocytoma before any adrenalectomy.

Surgery for Metabolic Syndrome

- **Adrenocorticotrophic hormone (ACTH)-independent Cushing syndrome** from a functional tumour → abdominal obesity, hyperglycaemia, dyslipidaemia, hypertension; **insulin resistance rises and insulin secretion falls**.
- **Dyslipidaemia**: high very-low-density lipoprotein (VLDL), low-density lipoprotein (LDL), triglycerides and total cholesterol; low high-density lipoprotein (HDL); hepatic steatosis.
- **After adrenalectomy the response is variable**: insulin resistance and impaired secretion may persist; complete resolution also described.

Surgery for Systemic Diseases

- **Congenital adrenal hyperplasia (CAH)**: loss of a steroidogenic enzyme, **overwhelmingly 21-hydroxylase** → impaired cortisol and some aldosterone → precursors shunted to androgens.
- **Medical therapy since the 1950s**: glucocorticoid ± mineralocorticoid; supraphysiological doses needed to suppress the hypothalamic–pituitary axis → **iatrogenic Cushing syndrome**.
- **Bilateral adrenalectomy**: deliberate adrenal insufficiency with fixed replacement — reproducible, avoids Cushing syndrome. **Concerns: adrenal crisis; gonadal (ectopic) adrenal rest tumours**.
- **Meta-analysis**: symptomatic improvement in most (hyperandrogenism, growth velocity, cushingoid features), with adrenal crisis and adrenal rest tumours in a minority.

PARTIAL ADRENALECTOMY

- **History**: 1918 animal model — survival without insufficiency after removal of up to seven-eighths of the gland. **1934 — first human**: subtotal bilateral adrenalectomy for hyperadrenalism via staged dorsal lumbotomies, no mortality. **First partial adrenalectomy for tumour**: performed 1965, reported 1983 — bilateral pheochromocytoma, right total and left partial; off steroids 15 years later with a mass-free remnant.
- **Indications**: children; **bilateral pheochromocytoma**; Conn syndrome; non-functioning adenomas with severe hypertension; adrenal cysts; **familial syndromes**; **prior contralateral adrenalectomy**.

Surgical Technique

- **Approach as for the respective side;** only the resection differs.
- **Arteries enter inferiorly, medially and superiorly → lateral dissection is safe;** elevating the lateral portion aids exposure.
- **Preserve the arterial supply; the main adrenal vein can be divided** — rich collateral venous drainage maintains remnant outflow (important in partial adrenalectomy for pheochromocytoma).
- **Margins: intraoperative US; indocyanine green (ICG)** to assess remnant viability.

TISSUE DIVISION IN PARTIAL ADRENALECTOMY

Method	Feature
Sharp division between metal or locking clips	Clean margin, good visualisation throughout
Bipolar diathermy	Protects surrounding tissue from heat; any platform
Thin stapler	Seals and divides; bulky, cumbersome around tissue
Impedance-sensing sealing devices	Seal then cut; tissue impedance signals when division is safe
Topical haemostatic agents	Liquid, gel, powder, cloth — residual ooze

- **MIS: bag the specimen** to prevent rupture and spill.

Concept — *partial adrenalectomy works because arterial inflow enters from three sides while venous outflow is collateralised — preserve the arteries and the main vein becomes expendable.*

OPEN INCISIONS

- **Superior exposure is the main challenge.** The left is accessible through most incisions; the right is harder — more cephalad and with its caval venous drainage.

OPEN INCISIONS COMPARED

Incision	Position and line	Strengths	Drawbacks / best for
Midline (laparotomy)	Supine; vertical through linea alba (Fig. 147.9)	No muscle divided, few crossing nerves; reproducible; whole abdomen and RP; easy vascular control	Upper RP hard without mobilising structures; large masses
Subcostal / chevron	Supine, slight bump; tip of 11th rib toward xiphoid, 1–2 cm below costal margin; extend across midline → chevron	Direct great-vessel access; excellent upper-abdominal and RP exposure	More pain (regional anaesthesia helps); large tumours, vascular or ipsilateral renal involvement; chevron for bilateral or large unilateral
Thoracoabdominal (Ivor Lewis)	Body roll under operative flank; arm and shoulder rotated away in a sling (Fig. 147.10); joins pleural and peritoneal cavities	Unparalleled exposure	Highest incision-specific morbidity; rarely used — large tumours or IVC extension above the liver
Flank	45° lateral, flexed above iliac crest; above 11th or 12th rib toward umbilicus ± partial rib removal	Familiar renal approach; extensible to lateral rectus border	Superior dissection must be allowed for — often 11th rib

Incision	Position and line	Strengths	Drawbacks / best for
Posterior (prone lumbodorsal)	Prone	Most direct route ; bilateral small tumours without repositioning	Limited exposure and vascular control; small functional tumours , not large or malignant

- **Subcostal layers:** anterior rectus sheath, rectus, external oblique, internal oblique, transversus abdominis, transversalis fascia, peritoneum.
- **Flank sequence:** skin, subcutaneous fat, Scarpa's fascia → partially divide latissimus dorsi laterally and external oblique medially to expose rib → if sparing rib, **dissect on its superior border** (bundle in posteroinferior groove) → internal oblique, transversus → open transversalis fascia → sweep peritoneum anteromedially.

Concept — *incision choice trades exposure against morbidity along one axis — from the posterior approach (most direct, least exposure) to the thoracoabdominal (most exposure, most morbidity) — with tumour size and suspicion of malignancy deciding how far along it to go.*

ABLATION OF THE ADRENAL

- **Ancient principle:** Hippocrates held that what the knife cannot cure, fire cures. Most solid-organ ablation data come from liver, extended to prostate, kidney and now adrenal.
- **Routes:** transcutaneous, percutaneous, laparoscopic; mostly by interventional radiology, but urologists must know the technology. **Biopsy before or at ablation.**

Energy	Modalities	Adrenal status
Heat	High-intensity focused ultrasound (HIFU), microwave, radiofrequency ablation (RFA) , laser	RFA most useful: focused, little collateral effect, predictable destruction; others in small non-randomised series
Cold	Cryotherapy	Extensive in kidney; very limited in adrenal

RFA mechanism: generator alternating current → electrode with uninsulated tip placed under US or CT → **ionic agitation** around tip → frictional heat **>50–60 °C** → protein denaturation, **coagulative necrosis**. Cooled-saline tip protects normal tissue and improves tumour conductivity; **grounding pad required**.

- **Ablation of pheochromocytoma or ACC is highly discouraged** outside a stringent research protocol.

FUTURE OF ADRENAL SURGERY

- **Robotics, big data, AI, genomics, virtual reality and AR with improved imaging** — tools to strengthen care, never replacements for decision-making and technique.
- **AI in surgery is in its infancy:** preoperative planning, intraoperative assistance, robotics. More used in orthopaedics and neurosurgery; abdominal barriers are patient diversity, complex and changing anatomy, and multimodal input. Reconstruction of the patient's anatomy allows simulated rehearsal.
- **Digital surgery:** technology enhancing preoperative planning, surgical performance, therapeutic support or training to improve outcomes and minimise harm — already present (tissue discrimination, landmark identification, video review).
- **Most new urologists train and operate robotically;** open surgery keeps a place by pathology; surgeons should shape next-generation platform design.